be implemented is the visualization of the data that have been collected so far, the current GP belief, and how a BayesOpt policy is scoring the rest of the data points. The target of this visualization is shown in Figure 4.3, which we saw in Chapter 1.

Specifically, the top panel of the plot shows the predictions made by a GP model against the true objective function, while the bottom panel shows the acquisition scores as computed by a BayesOpt policy.

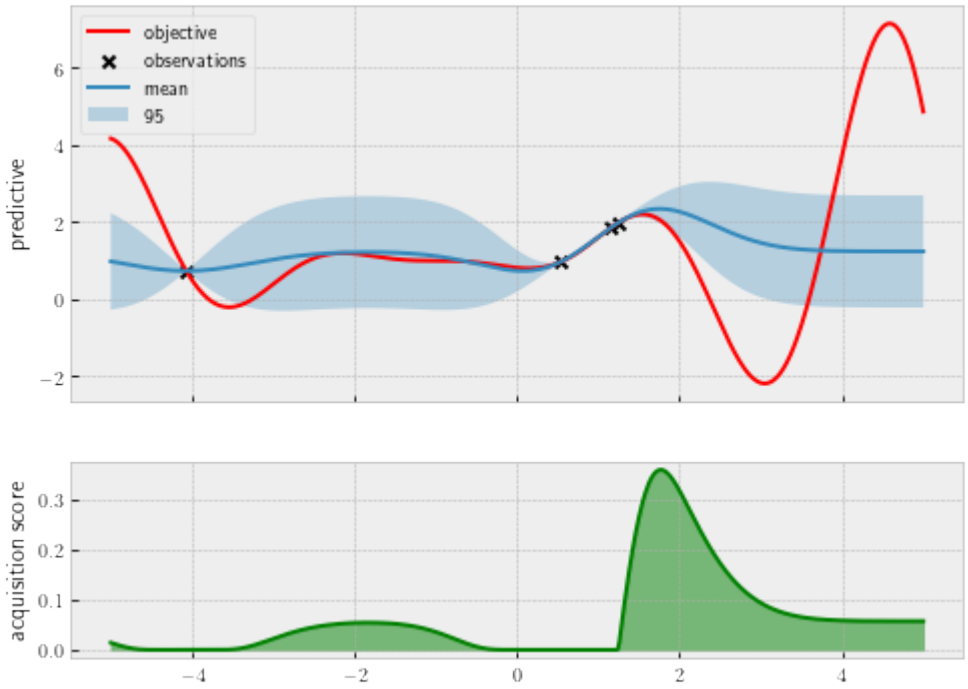


Figure 4.3 A typical visualization of progress in Bayesian optimization. The top panel shows the GP predictions and the ground truth objective function, while the bottom panel shows the acquisition scores made by a Bayesian optimization policy named Expected Improvement, which we will learn in Section 4.3.

We are already familiar with how to generate the top panel, and generating the bottom panel is just as simple. This will be done using a helper function similar to the one we used in Section 3.3. The function takes in a GP model and its likelihood function, as well as two optional inputs:

1. `policy` refers to a BayesOpt policy object, which can be called in a similar way as any PyTorch module. Here, we are calling it on the grid `xs` that represents our search space to obtain the acquisition scores across the space. We will be seeing how to implement these policy objects with BoTorch in the next section, but we don't need to know more about these objects right now.
2. `next_x` is the location of the data point that maximizes the acquisition score, which is to be added to the running training data.

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